Extension Justification

Clio Selyuzhitsky

1 Extension Description

I perform a robustness check on "When the Levee Breaks: Black Migration and Economic Development in the American South" by Richard Hornbeck and Suresh Naidu" by questioning the use of lags. In the paper, reasonably, when analyzing the impact of flooding on capital, they include lags for 4 time periods prior, which is about 20 years. Since time scales in the paper are very large, it makes econometrically, and since individuals need to save money in order to acquire capital, it makes sense economically. However, 20 years feels arbitrary to me. In my replication, I test the robustness of their regression by slowly increasing the number of lags from none to fully specified as in the paper. I always include controls. In other words, my regression is:

$$Y_{ct} = \beta_t Flood_c + \alpha_{st} + \alpha_c + \theta_t X_c + \sum_{\ell=1}^{L} \gamma_{t-\ell-1} Flood_c + \varepsilon_{ct}$$

for $L \in \{1, 2, 3, 4, 5\}$. As far as I can tell, in the paper, the include these lags in β_t without explaining in the specification.

2 Extension results

In general, the standard errors largely did not change between numbers of lags. On the other hand, going between no lags to including a single lag noticeably increased equipment values estimates, and decreased farm size estimates. There was also another decreasing jump when fully specifying the model, but only for farm size. Significance does not change across regressions at all. The tables are listed below.

Table 1: Farm Size Regressions

	(1) *no lag*	(2) *1 lag*	(3) *2 lags*	(4) *3 lags*	(5) *fully specified*
f_int_1930	0.137*	0.082	0.069	0.068	0.042
	(0.068)	(0.071)	(0.062)	(0.063)	(0.064)
f_int_1935	0.316**	0.265**	0.249**	0.246**	0.187**
	(0.071)	(0.076)	(0.065)	(0.067)	(0.062)
f_int_1940	0.322**	0.226*	0.210**	0.206**	0.171*
	(0.085)	(0.087)	(0.078)	(0.078)	(0.083)
f_int_1945	0.450**	0.347**	0.321**	0.313**	0.242**
	(0.091)	(0.098)	(0.077)	(0.080)	(0.077)
f_int_1950	0.606**	0.529**	0.506**	0.492**	0.416**
	(0.099)	(0.106)	(0.091)	(0.092)	(0.091)
f_int_1954	0.754**	0.657**	0.631**	0.618**	0.536**
	(0.114)	(0.122)	(0.106)	(0.109)	(0.107)
f_int_1960	1.047**	0.857**	0.840**	0.822**	0.719**
	(0.146)	(0.148)	(0.139)	(0.142)	(0.142)
f_int_1964	1.307**	1.044**	1.023**	1.004**	0.891**
	(0.172)	(0.173)	(0.161)	(0.167)	(0.166)
f_int_1970	1.122**	0.779**	0.776**	0.755**	0.644**
	(0.167)	(0.161)	(0.159)	(0.165)	(0.162)
N	1630	1630	1630	1630	1630

Standard errors in parentheses.
* $p<0.05$, ** $p<0.01$

Table 2: Equipment Value Regressions

	(6)	(7)	(8)	(9)	(10)
	no lag	*1 lag*	*2 lags*	*3 lags*	*fully specified*
f_int_1930	-0.191	-0.152	-0.103	-0.103	-0.101
	(0.137)	(0.119)	(0.111)	(0.113)	(0.113)
f_int_1935	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)
f_int_1940	0.253*	0.279*	0.318**	0.325**	0.326**
	(0.117)	(0.120)	(0.119)	(0.121)	(0.124)
f_int_1945	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)
f_int_1950	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)
f_int_1954	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)
f_int_1960	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)
f_int_1964	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)
f_int_1970	0.583**	0.610**	0.653**	0.672**	0.673**
	(0.162)	(0.162)	(0.159)	(0.165)	(0.166)
N	652	652	652	652	652

Standard errors in parentheses.
* $p<0.05$, ** $p<0.01$